Table 1: Specifications of the Mixer architectures. The "B", "L", and "H" (base, large, and huge) model scales follow Dosovitskiy et al. [14]. A brief notation "B/16" means the model of base scale with patches of resolution 16×16. The number of parameters is reported for an input resolution of 224 and does not include the weights of the classifier head.

Specification	S/32	S/16	B/32	B/16	L/32	L/16	H/14
Number of layers	8	8	12	12	24	24	32
Patch resolution $P \times P$	32×32	16×16	32×32	16×16	32×32	16×16	14×14
Hidden size C	512	512	768	768	1024	1024	1280
Sequence length S	49	196	49	196	49	196	256
MLP dimension D_C	2048	2048	3072	3072	4096	4096	5120
MLP dimension D_S	256	256	384	384	512	512	640
Parameters (M)	19	18	60	59	206	207	431

dropout [43], and stochastic depth [19]. This set of techniques was inspired by the *timm library* [54] and Touvron et al. [48]. More details on these hyperparameters are provided in Supplementary B.

Fine-tuning We fine-tune using momentum SGD, batch size 512, gradient clipping at global norm 1, and a cosine learning rate schedule with a linear warmup. We do not use weight decay when fine-tuning. Following common practice [22, 48], we also fine-tune at higher resolutions with respect to those used during pre-training. Since we keep the patch resolution fixed, this increases the number of input patches (say from S to S') and thus requires modifying the shape of Mixer's token-mixing MLP blocks. Formally, the input in Eq. (1) is left-multiplied by a weight matrix $\mathbf{W}_1 \in \mathbb{R}^{D_S \times S}$ and this operation has to be adjusted when changing the input dimension S. For this, we increase the hidden layer width from D_S to $D_{S'}$ in proportion to the number of patches and initialize the (now larger) weight matrix $\mathbf{W}_2' \in \mathbb{R}^{D_{S'} \times S'}$ with a block-diagonal matrix containing copies of $\mathbf{W}_2$ on its diagonal. This particular scheme only allows for $S' = K^2 S$ with $K \in \mathbb{N}$. See Supplementary C for further details. On the VTAB-1k benchmark we follow the BiT-HyperRule [22] and fine-tune Mixer models at resolution 224 and 448 on the datasets with small and large input images respectively.

Metrics We evaluate the trade-off between the model's computational cost and quality. For the former we compute two metrics: (1) Total pre-training time on TPU-v3 accelerators, which combines three relevant factors: the theoretical FLOPs for each training setup, the computational efficiency on the relevant training hardware, and the data efficiency. (2) Throughput in images/sec/core on TPU-v3. Since models of different sizes may benefit from different batch sizes, we sweep the batch sizes and report the highest throughput for each model. For model quality, we focus on top-1 downstream accuracy after fine-tuning. On two occasions (Figure 3, right and Figure 4), where fine-tuning all of the models is too costly, we report the few-shot accuracies obtained by solving the ℓ_2-regularized linear regression problem between the frozen learned representations of images and the labels.

Models We compare various configurations of Mixer, summarized in Table 1, to the most recent, state-of-the-art, CNNs and attention-based models. In all the figures and tables, the MLP-based Mixer models are marked with pink (●), convolution-based models with yellow (●), and attention-based models with blue (●). The Vision Transformers (ViTs) have model scales and patch resolutions similar to Mixer. HaloNets are attention-based models that use a ResNet-like structure with local self-attention layers instead of 3×3 convolutions [51]. We focus on the particularly efficient "HaloNet-H4 (base 128, Conv-12)" model, which is a hybrid variant of the wider HaloNet-H4 architecture with some of the self-attention layers replaced by convolutions. Note, we mark HaloNets with both attention and convolutions with blue (●). Big Transfer (BiT) [22] models are ResNets optimized for transfer learning. NFNets [7] are normalizer-free ResNets with several optimizations for ImageNet classification. We consider the NFNet-F4+ model variant. We consider MPL [34] and ALIGN [21] for EfficientNet architectures. MPL is pre-trained at very large-scale on JFT-300M images, using meta-pseudo labelling from ImageNet instead of the original labels. We compare to the EfficientNet-B6-Wide model variant. ALIGN pre-train image encoder and language encoder on noisy web image text pairs in a contrastive way. We compare to their best EfficientNet-L2 image encoder.

3.1 Main results

Table 2 presents comparison of the largest Mixer models to state-of-the-art models from the literature. "ImNet" and "ReaL" columns refer to the original ImageNet validation [13] and cleaned-up ReaL [5]

Table 2: Transfer performance, inference throughput, and training cost. The rows are sorted by inference throughput (fifth column). Mixer has comparable transfer accuracy to state-of-the-art models with similar cost. The Mixer models are fine-tuned at resolution 448. Mixer performance numbers are averaged over three fine-tuning runs and standard deviations are smaller than 0.1.

	ImNet top-1	ReaL top-1	Avg 5 top-1	VTAB-1k 19 tasks	Throughput img/sec/core	TPUv3 core-days
Pre-trained on ImageNet-21k (public)						
● HaloNet [51]	85.8	—	—	—	120	0.10k
● Mixer-L/16	84.15	87.86	93.91	74.95	105	0.41k
● ViT-L/16 [14]	85.30	88.62	94.39	72.72	32	0.18k
● BiT-R152x4 [22]	85.39	—	94.04	70.64	26	0.94k
Pre-trained on JFT-300M (proprietary)						
● NFNet-F4+ [7]	89.2	—	—	—	46	1.86k
● Mixer-H/14	87.94	90.18	95.71	75.33	40	1.01k
● BiT-R152x4 [22]	87.54	90.54	95.33	76.29	26	9.90k
● ViT-H/14 [14]	88.55	90.72	95.97	77.63	15	2.30k
Pre-trained on unlabelled or weakly labelled data (proprietary)						
● MPL [34]	90.0	91.12	—	—	—	20.48k
● ALIGN [21]	88.64	—	—	79.99	15	14.82k

labels. "Avg. 5" stands for the average performance across all five downstream tasks (ImageNet, CIFAR-10, CIFAR-100, Pets, Flowers). Figure 2 (left) visualizes the accuracy-compute frontier. When pre-trained on ImageNet-21k with additional regularization, Mixer achieves an overall strong performance (84.15% top-1 on ImageNet), although slightly inferior to other models[2]. Regularization in this scenario is necessary and Mixer overfits without it, which is consistent with similar observations for ViT [14]. The same conclusion holds when training Mixer from random initialization on ImageNet (see Section 3.2): Mixer-B/16 attains a reasonable score of 76.4% at resolution 224, but tends to overfit. This score is similar to a vanilla ResNet50, but behind state-of-the-art CNNs/hybrids for the ImageNet "from scratch" setting, e.g. 84.7% BotNet [42] and 86.5% NFNet [7].

When the size of the upstream dataset increases, Mixer's performance improves significantly. In particular, Mixer-H/14 achieves 87.94% top-1 accuracy on ImageNet, which is 0.5% better than BiT-ResNet152x4 and only 0.5% lower than ViT-H/14. Remarkably, Mixer-H/14 runs 2.5 times faster than ViT-H/14 and almost twice as fast as BiT. Overall, Figure 2 (left) supports our main claim that in terms of the accuracy-compute trade-off Mixer is competitive with more conventional neural network architectures. The figure also demonstrates a clear correlation between the total pre-training cost and the downstream accuracy, even across architecture classes.

BiT-ResNet152x4 in the table are pre-trained using SGD with momentum and a long schedule. Since Adam tends to converge faster, we complete the picture in Figure 2 (left) with the BiT-R200x3 model from Dosovitskiy et al. [14] pre-trained on JFT-300M using Adam. This ResNet has a slightly lower accuracy, but considerably lower pre-training compute. Finally, the results of smaller ViT-L/16 and Mixer-L/16 models are also reported in this figure.

3.2 The role of the model scale

The results outlined in the previous section focus on (large) models at the upper end of the compute spectrum. We now turn our attention to smaller Mixer models.

We may scale the model in two independent ways: (1) Increasing the model size (number of layers, hidden dimension, MLP widths) when pre-training; (2) Increasing the input image resolution when

[2]In Table 2 we consider the highest accuracy models in each class for each pre-training dataset. These all use the large resolutions (448 and above). However, fine-tuning at smaller resolution can lead to substantial improvements in the test-time throughput, with often only a small accuracy penalty. For instance, when pre-training on ImageNet-21k, the Mixer-L/16 model fine-tuned at 224 resolution achieves 82.84% ImageNet top-1 accuracy at throughput 420 img/sec/core; the ViT-L/16 model fine-tuned at 384 resolution achieves 85.15% at 80 img/sec/core [14]; and HaloNet fine-tuned at 384 resolution achieves 85.5% at 258 img/sec/core [51].

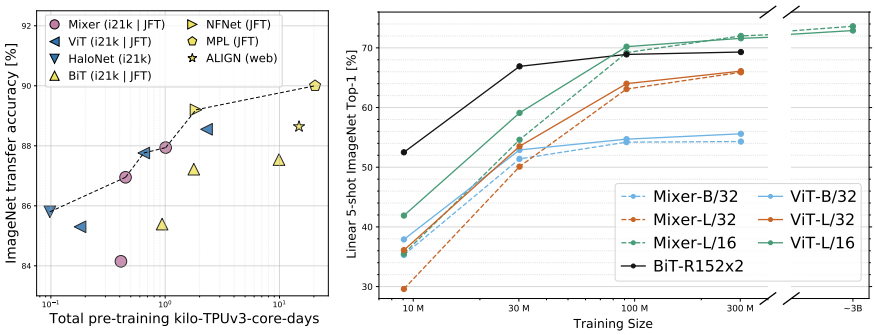

Figure 2: **Left:** ImageNet accuracy/training cost Pareto frontier (dashed line) for the SOTA models in Table 2. Models are pre-trained on ImageNet-21k, or JFT (labelled, or pseudo-labelled for MPL), or web image text pairs. Mixer is as good as these extremely performant ResNets, ViTs, and hybrid models, and sits on frontier with HaloNet, ViT, NFNet, and MPL. **Right:** Mixer (solid) catches or exceeds BiT (dotted) and ViT (dashed) as the data size grows. Every point on a curve uses the same pre-training compute; they correspond to pre-training on 3%, 10%, 30%, and 100% of JFT-300M for 233, 70, 23, and 7 epochs, respectively. Additional points at $\sim$3B correspond to pre-training on an even larger JFT-3B dataset for the same number of total steps. Mixer improves more rapidly with data than ResNets, or even ViT. The gap between large Mixer and ViT models shrinks.

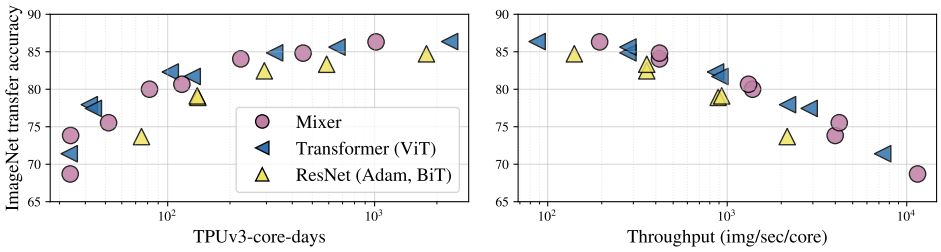

Figure 3: The role of the model scale. ImageNet validation top-1 accuracy vs. total pre-training compute (**left**) and throughput (**right**) of ViT, BiT, and Mixer models at various scales. All models are pre-trained on JFT-300M and fine-tuned at resolution 224, which is lower than in Figure 2 (left).

fine-tuning. While the former affects both pre-training compute and test-time throughput, the latter only affects the throughput. Unless stated otherwise, we fine-tune at resolution 224.

We compare various configurations of Mixer (see Table 1) to ViT models of similar scales and BiT models pre-trained with Adam. The results are summarized in Table 3 and Figure 3. When trained from scratch on ImageNet, Mixer-B/16 achieves a reasonable top-1 accuracy of 76.44%. This is 3% behind the ViT-B/16 model. The training curves (not reported) reveal that both models achieve very similar values of the training loss. In other words, Mixer-B/16 overfits more than ViT-B/16. For the Mixer-L/16 and ViT-L/16 models this difference is even more pronounced.

As the pre-training dataset grows, Mixer's performance steadily improves. Remarkably, Mixer-H/14 pre-trained on JFT-300M and fine-tuned at 224 resolution is only 0.3% behind ViT-H/14 on ImageNet whilst running 2.2 times faster. Figure 3 clearly demonstrates that although Mixer is slightly below the frontier on the lower end of model scales, it sits confidently on the frontier at the high end.

3.3 The role of the pre-training dataset size

The results presented thus far demonstrate that pre-training on larger datasets significantly improves Mixer's performance. Here, we study this effect in more detail.

To study Mixer's ability to make use of the growing number of training examples we pre-train Mixer-B/32, Mixer-L/32, and Mixer-L/16 models on random subsets of JFT-300M containing 3%, 10%, 30% and 100% of all the training examples for 233, 70, 23, and 7 epochs. Thus, every model is pre-trained for the same number of total steps. We also pre-train Mixer-L/16 model on an even larger JFT-3B dataset [59] containing roughly 3B images with 30k classes for the same number of total steps.